

MODULE - 03

Z - TRANSFORMS

①
L.T. to solve continuous
Z.T. to solve discrete systems

- * Z-transform operates on the sequence of functions of a single variable defined for non negative integral values of the variable. This transform has number of properties similar to that of Laplace Transform.
- * *Z.T. is a tool to solve difference Eq. & to solve discrete systems.*
Difference Equations arises in situations with the data consisting of only a set of values of an unknown function (discrete values).
Just as Laplace Transform and Fourier Transform serves as a tool to solve some types of ordinary and partial differential equations, Z transform serves as a tool to solve difference equations.
- * Z-transform play an important role in the analysis & representations of discrete-time linear shift invariant systems. It has applications in control system of engineering & also in some advanced statistical problems.
- * Applications.
 - Z transform used in creating audio effects for digital systems or in designing radar signal processors.
 - It is primarily used to analyze & process digital data.
 - It converts a discrete-time signal, which is sequence of real or complex numbers, into a complex valued frequency domain representation.

* Definition of Z-transform

If $u_n = f(n)$ defined for all $n = 0, 1, 2, 3, \dots$ & $u_n = 0$ for $n < 0$ then the Z-transform of u_n denoted by $Z_T(u_n)$ is defined by

$$\boxed{Z_T(u_n) = \sum_{n=0}^{\infty} u_n z^{-n}} \quad \text{--- (1)}$$

whenever the series on the RHS of (1) converges, it will be a function of z & we write,

$$Z_T(u_n) = \bar{u}(z) \quad \text{--- (2)}$$

Further (2) can be written in the equivalent form

$$\boxed{z^{-1} [\bar{u}(z)] = u_n} \quad \text{--- (3)}$$

This is called the Inverse Z-transform.

* Z-Transform exists only when the infinite series in (1) is convergent.

* Note:

notation $Z(u_n)$, $z^{-1} [\bar{u}(z)]$ is also used.

notation $U(z)$ for $\bar{u}(z)$ is also used.

* Property: $Z_T(n^k) = -z \frac{d}{dz} Z_T(n^{k-1})$ where k is a +ve integer.

proof: $\rightarrow$ consider RHS

$$\begin{aligned} \text{i.e., } -z \frac{d}{dz} Z_T(n^{k-1}) &= -z \frac{d}{dz} \sum_{n=0}^{\infty} n^{k-1} z^{-n} \\ &= -z \sum_{n=0}^{\infty} n^{k-1} (-n) z^{-n-1} \\ &= \sum_{n=0}^{\infty} n^k z^{-n} = Z_T(n^k) = \text{LHS} \end{aligned}$$

Thus we have proved that, $Z_T(n^k) = -z \frac{d}{dz} Z_T(n^{k-1})$

* Z-transform of some standard functions [standard Z transform] (3)

① $Z_T(k^n)$

② $Z_T(1)$

③ $Z_T(n)$

④ $Z_T(n^2)$

⑤ $Z_T(n^3)$

① $Z_T(k^n)$

By the definition, $Z_T(k^n) = \sum_{n=0}^{\infty} k^n z^{-n} = \sum_{n=0}^{\infty} (k/z)^n$
 $= 1 + (k/z) + (k/z)^2 + \dots$

The series on the RHS is a geometric series of the form $1 + r + r^2 + \dots$ whose sum to infinity is $1/(1-r)$ where $r = k/z$.

$$\therefore Z_T(k^n) = \frac{1}{1 - (k/z)}$$

Thus, $Z_T(k^n) = \frac{z}{z - k}$

② $Z_T(1)$

Putting $k = 1$ in the previous result we have $k^n = 1$

Thus $Z_T(1) = \frac{z}{z - 1}$

Remark: The result can easily be established independently also.

Results (3), (4), (5) are established by using the property, $Z_T(n^k) = -z \frac{d}{dz} Z_T(n^{k-1})$

we obtain the required Z-transforms by taking $k=1$, $k=2$, $k=3$.

(3) $Z_T(n)$

$$\text{when } k=1, Z_T(n) = -z \frac{d}{dz} Z_T(n^0) = -z \frac{d}{dz} Z_T(1)$$

$$\text{i.e. } Z_T(n) = -z \frac{d}{dz} \left[\frac{z}{z-1} \right] = -z \left[\frac{(z-1) - z}{(z-1)^2} \right]$$

$$\text{Thus, } Z_T(n) = \frac{z}{(z-1)^2}$$

(4) $Z_T(n^2)$

$$\text{when } k=2, Z_T(n^2) = -z \frac{d}{dz} Z_T(n)$$

$$\text{i.e. } Z_T(n^2) = -z \frac{d}{dz} \left[\frac{z}{(z-1)^2} \right]$$

$$= -z \left\{ \frac{(z-1)^2 - z \cdot 2(z-1)}{(z-1)^4} \right\}$$

$$= -z(z-1) \left\{ \frac{z-1-2z}{(z-1)^4} \right\}$$

$$= \frac{-z(-z-1)}{(z-1)^3}$$

$$\text{Thus, } Z_T(n^2) = \frac{z(z+1)}{(z-1)^3} = \frac{z^2+z}{(z-1)^3}$$

(5) $Z_T(n^3)$

$$\text{when } k=3, Z_T(n^3) = -z \frac{d}{dz} Z_T(n^2)$$

i.e.,

$$Z_T(n^3) = -z \frac{d}{dz} \left[\frac{z^2+z}{(z-1)^3} \right]$$

$$Z_T(n^3) = -z \left\{ \frac{(z-1)^3(2z+1) - (z^2+z) \cdot 3(z-1)^2}{(z-1)^6} \right\}$$

$$= -z (z-1)^2 \left\{ \frac{(z-1)(2z+1) - 3(z^2+z)}{(z-1)^4} \right\}$$

$$Z_T(n^3) = \frac{-z(-z^2-4z-1)}{(z-1)^4} = \frac{z^3+4z^2+z}{(z-1)^4}$$

$$\text{Thus, } Z_T(n^3) = \frac{z^3+4z^2+z}{(z-1)^4}$$

* PROPERTIES

① LINEARITY PROPERTY

If u_n & v_n be any two discrete valued functions then
 $Z_T(c_1 u_n + c_2 v_n) = c_1 Z_T(u_n) + c_2 Z_T(v_n)$
where c_1, c_2 are constants.

→ we have by the definition,

$$\begin{aligned} Z_T(c_1 u_n + c_2 v_n) &= \sum_{n=0}^{\infty} (c_1 u_n + c_2 v_n) z^{-n} \\ &= c_1 \sum_{n=0}^{\infty} u_n z^{-n} + c_2 \sum_{n=0}^{\infty} v_n z^{-n} \end{aligned}$$

$$\text{Thus, } \boxed{Z_T(c_1 u_n + c_2 v_n) = c_1 Z_T(u_n) + c_2 Z_T(v_n)}$$

⑤ DAMPING RULE (PROPERTY)

⑥

If $Z_T(u_n) = \bar{u}(z)$ then

$$\textcircled{i} \quad Z_T(k^n u_n) = \bar{u}(z/k)$$

$$\textcircled{ii} \quad Z_T(k^{-n} u_n) = \bar{u}(kz)$$

$$\rightarrow \textcircled{i} \quad Z_T(k^n u_n) = \sum_{n=0}^{\infty} (k^n u_n) z^{-n}$$

$$Z_T[k^n u_n] = \sum_{n=0}^{\infty} u_n (z/k)^{-n} = \bar{u}(z/k)$$

$$\therefore \text{Thus, } \boxed{Z_T(k^n u_n) = \bar{u}(z/k)}$$

$$\textcircled{ii} \quad Z_T(k^{-n} u_n) = \sum_{n=0}^{\infty} (k^{-n} u_n) z^{-n}$$

$$= \sum_{n=0}^{\infty} u_n (kz)^{-n}$$

$$= \bar{u}(kz)$$

$$\therefore \text{Thus, } \boxed{Z_T(k^{-n} u_n) = \bar{u}(kz)}$$

* Some applications of damping rule

we have already obtained $Z_T(n)$, $Z_T(n^2)$, $Z_T(n^3)$.

& we can obtain $Z_T(k^n n)$, $Z_T(k^n n^2)$, $Z_T(k^n n^3)$ by using damping rule.

$$\textcircled{i} \quad Z_T(k^n n) = \left\{ Z_T(n) \right\}_{z \rightarrow (z/k)} \quad \text{where } Z_T(n) = \frac{z}{(z-1)^2}$$

$$\therefore Z_T(k^n n) = \frac{(z/k)}{(z/k-1)^2}$$

Thus, $Z_T(k^n u_n) = \frac{kz}{(z-k)^2}$ (4)

(ii) we have, $Z_T(n^2) = \frac{z^2 + z}{(z-1)^3}$

$\therefore Z_T(k^n n^2) = \left\{ Z_T(n^2) \right\}_{z \rightarrow (z/k)} = \frac{(z/k)^2 + (z/k)}{(z/k-1)^3}$

Thus, $Z_T(k^n n^2) = \frac{kz^2 + k^2 z}{(z-k)^3}$

(iii) we have, $Z_T(n^3) = \frac{z^3 + 4z^2 + z}{(z-1)^4}$

$\therefore Z_T(k^n n^3) = \left\{ Z_T(n^3) \right\}_{z \rightarrow (z/k)} = \frac{(z/k)^3 + 4(z/k)^2 + (z/k)}{(z/k-1)^4}$

Thus, $Z_T(k^n n^3) = \frac{kz^3 + 4k^2 z^2 + k^3 z}{(z-k)^4}$

* LIST OF standard Z-transforms

(1) $Z_T(1) = \frac{z}{z-1}$

(6) $Z_T(k^n n^2) = \frac{kz^2 + k^2 z}{(z-k)^3}$

(2) $Z_T(k^n) = \frac{z}{z-k}$

(4) $Z_T(n^3) = \frac{z^3 + 4z^2 + z}{(z-1)^4}$

(3) $Z_T(n) = \frac{z}{(z-1)^2}$

(8) $Z_T(k^n n^3) = \frac{kz^3 + 4k^2 z^2 + k^3 z}{(z-k)^4}$

(4) $Z_T(k^n n) = \frac{kz}{(z-k)^2}$

(5) $Z_T(n^2) = \frac{z^2 + z}{(z-1)^3}$

③ SHIFTING RULE (property)

⑧

① Right shifting rule

If $Z_T(u_n) = \bar{u}(z)$ then $Z_T(u_{n-k}) = z^{-k} \bar{u}(z)$
where $k > 0$.

→ By the definition,

$$Z_T(u_{n-k}) = \sum_{n=0}^{\infty} u_{n-k} z^{-n}$$

Since $u_n = 0$ for $n < 0$ in the general context,
we have $u_{n-k} = 0$ for $n = 0, 1, 2, \dots, (k-1)$.

$$\begin{aligned} \therefore Z_T(u_{n-k}) &= \sum_{n=k}^{\infty} u_{n-k} z^{-n} \\ &= u_0 z^{-k} + u_1 z^{-(k+1)} + u_2 z^{-(k+2)} + \dots \end{aligned}$$

$$\text{i.e., } = z^{-k} (u_0 + u_1 z^{-1} + u_2 z^{-2} + \dots)$$

$$= z^{-k} \sum_{n=0}^{\infty} u_n z^{-n} = z^{-k} \bar{u}(z)$$

$$\text{Then, } Z_T(u_{n-k}) = z^{-k} \bar{u}(z)$$

② Left Shifting rule

$$Z_T(u_{n+k}) = z^k \left[\bar{u}(z) - u_0 - u_1 z^{-1} - u_2 z^{-2} - \dots - u_{k-1} z^{-(k-1)} \right]$$

$$Z_T(u_{n+k}) = z^k \left[\bar{u}(z) - \sum_{r=0}^{k-1} u_r z^{-r} \right]$$

$$\rightarrow Z_T(u_{n+k}) = \sum_{n=0}^{\infty} u_{n+k} z^{-n} = z^k \sum_{n=0}^{\infty} u_{n+k} z^{-(n+k)}$$

(9)

$$\begin{aligned}
 \text{i.e., } Z_T(u_{n+k}) &= z^k \left[u_k z^{-k} + u_{k+1} z^{-(k+1)} + u_{k+2} z^{-(k+2)} + \dots \right] \\
 &= z^k \left[(u_0 + u_1 z^{-1} + u_2 z^{-2} + \dots + u_{k-1} z^{-(k-1)} + u_k z^{-k} \right. \\
 &\quad \left. + u_{k+1} z^{-(k+1)} + \dots) - \right. \\
 &\quad \left. (u_0 + u_1 z^{-1} + u_2 z^{-2} + \dots + u_{k-1} z^{-(k-1)}) \right] \\
 &= z^k \left[\sum_{n=0}^{\infty} u_n z^{-n} - (u_0 + u_1 z^{-1} + u_2 z^{-2} + \dots + u_{k-1} z^{-(k-1)}) \right]
 \end{aligned}$$

$$\text{Thus, } Z_T(u_{n+k}) = z^k \left[\bar{u}(z) - u_0 - u_1 z^{-1} - u_2 z^{-2} - \dots - u_{k-1} z^{-(k-1)} \right]$$

* note:- some particular cases.

- ① $Z_T(u_{n+1}) = z (\bar{u}(z) - u_0)$
- ② $Z_T(u_{n+2}) = z^2 [\bar{u}(z) - u_0 - u_1 z^{-1}]$
- ③ $Z_T(u_{n+3}) = z^3 [\bar{u}(z) - u_0 - u_1 z^{-1} - u_2 z^{-2}] \text{ etc.}$

* PROBLEMS

① Find the Z-transforms of the following.

- ① e^{-an} ② $e^{-an} \cdot n$ ③ $e^{-an} n^2$

→ ① $e^{-an} = (e^{-a})^n = k^n$ (say) where $k = e^{-a}$

we have, $Z_T(k^n) = \frac{z}{z-k}$

Thus, $Z_T(e^{-an}) = \frac{z}{z-e^{-a}}$

② $e^{-an} \cdot n$

we have $Z_T(n) = \frac{z}{(z-1)^2}$

$\therefore Z_T(k^n n) = \left\{ \frac{z}{(z-1)^2} \right\}_{z \rightarrow (z/k)} = \frac{(z/k)}{(z/k-1)^2}$

i.e., $Z_T(k^n n) = \frac{kz}{(z-k)^2}$

Thus by taking $k = e^{-a}$ we obtain,

$Z_T(e^{-an} \cdot n) = \frac{e^{-a} z}{(z-e^{-a})^2}$

③ we have, $Z_T(n^2) = \frac{z^2+z}{(z-1)^3}$

$\therefore Z_T(k^n n^2) = \left\{ \frac{z^2+z}{(z-1)^3} \right\}_{z \rightarrow (z/k)} = \frac{(z/k)^2 + (z/k)}{(z/k-1)^3}$

ie, $Z_T(k^n n^2) = \frac{kz^2 + k^2 z}{(z-k)^3} = \frac{kz(z+k)}{(z-k)^3}$

Thus by taking $k = e^{-a}$, we obtain

$$Z_T(e^{-an} \cdot n^2) = \frac{e^{-a} z(z+e^{-a})}{(z-e^{-a})^3}$$

② obtain the Z-transform of $\cos n\theta$ & $\sin n\theta$.

Hence deduce Z-transform of the following.

(i) $k^n \cos n\theta$ (ii) $k^n \sin n\theta$ (iii) $e^{-an} \cos n\theta$ (iv) $e^{-an} \sin n\theta$

→ w.b.t, $e^{in\theta} = \cos n\theta + i \sin n\theta$

we can write, $e^{in\theta} = (e^{i\theta})^n = k^n$ where $k = e^{i\theta}$

we have, $Z_T(k^n) = \frac{z}{(z-k)}$, k being $e^{i\theta}$

$$\begin{aligned} \therefore Z_T(e^{in\theta}) &= \frac{z}{z - e^{i\theta}} = \frac{z(z - e^{-i\theta})}{(z - e^{-i\theta})(z - e^{i\theta})} \\ &= \frac{z[z - (\cos\theta - i\sin\theta)]}{z^2 - z(e^{i\theta} + e^{-i\theta}) + 1} \\ &= \frac{z[(z - \cos\theta) + i\sin\theta]}{z^2 - 2z\cos\theta + 1} \end{aligned}$$

ie $Z_T(\cos n\theta + i \sin n\theta) = \frac{z[(z - \cos\theta) + i\sin\theta]}{z^2 - 2z\cos\theta + 1}$

(or) $Z_T(\cos n\theta) + i Z_T(\sin n\theta) = \left[\frac{z(z - \cos\theta)}{z^2 - 2z\cos\theta + 1} \right] + i \left[\frac{z\sin\theta}{z^2 - 2z\cos\theta + 1} \right]$

equating the real & imaginary parts we get

$$Z_T(\cos n\theta) = \frac{z(z - \cos\theta)}{z^2 - 2z\cos\theta + 1}$$

$$Z_T(\sin n\theta) = \frac{z \sin\theta}{z^2 - 2z\cos\theta + 1}$$

now, suppose $Z_T(\cos n\theta) = \bar{u}(z)$ &

$Z_T(\sin n\theta) = \bar{v}(z)$ then by damping rule,

$$Z_T(k^n \cos n\theta) = \bar{u}(z/k) \quad \&$$

$$Z_T(k^n \sin n\theta) = \bar{v}(z/k)$$

$$\therefore Z_T(k^n \cos n\theta) = \frac{(z/k)(z/k - \cos\theta)}{(z/k)^2 - 2 \cdot z/k \cdot \cos\theta + 1}$$

$$\text{Thus, } \boxed{Z_T(k^n \cos n\theta) = \frac{z(z - k \cos\theta)}{z^2 - 2kz \cos\theta + k^2}} \quad \text{--- (i)}$$

$$\text{Also, } Z_T(k^n \sin n\theta) = \frac{z/k \cdot \sin\theta}{(z/k)^2 - 2 \cdot z/k \cdot \cos\theta + 1}$$

$$\text{Thus, } \boxed{Z_T(k^n \sin n\theta) = \frac{kz \sin\theta}{z^2 - 2kz \cos\theta + k^2}} \quad \text{--- (ii)}$$

By taking $k = e^{-a}$ in (i) & (ii) we obtain the required (iii) & (iv) as follows.

$$\boxed{Z_T(e^{-an} \cos n\theta) = \frac{z(z - e^{-a} \cos\theta)}{z^2 - 2e^{-a}z \cos\theta + e^{-2a}}} \quad \text{--- (iii)}$$

$$\boxed{Z_T(e^{-an} \sin n\theta) = \frac{e^{-a}z \sin\theta}{z^2 - 2e^{-a}z \cos\theta + e^{-2a}}} \quad \text{--- (iv)}$$

3 ^{M.P.}

Find the Z-transform of $(n+1)^2$

$$\begin{aligned}
 \rightarrow Z_T[(n+1)^2] &= Z_T(n^2 + 2n + 1) \\
 &= Z_T(n^2) + 2Z_T(n) + Z_T(1) \\
 &= \frac{z^2 + z}{(z-1)^3} + 2 \cdot \frac{z}{(z-1)^2} + \frac{z}{z-1} \\
 &= \frac{z^2 + z + 2z(z-1) + z(z-1)^2}{(z-1)^3}
 \end{aligned}$$

Thus,

$$Z_T[(n+1)^2] = \frac{z^3 + z^2}{(z-1)^3}$$

4 ^{M.P.}

Obtain the Z-transform of $\cosh n\theta$ & $\sinh n\theta$.

$$\rightarrow \cosh n\theta = \frac{1}{2}(e^{n\theta} + e^{-n\theta}) = \frac{1}{2}\{(e^\theta)^n + (e^{-\theta})^n\}$$

$$\text{i.e., } \cosh n\theta = \frac{1}{2}\{p^n + q^n\} \text{ (say) where, } p = e^\theta \text{ \& } q = e^{-\theta}$$

$$\text{now, } Z_T(\cosh n\theta) = \frac{1}{2}\{Z_T(p^n) + Z_T(q^n)\}$$

$$= \frac{1}{2}\left\{\frac{z}{z-p} + \frac{z}{z-q}\right\}$$

$$= \frac{z}{2}\left\{\frac{1}{z-e^\theta} + \frac{1}{z-e^{-\theta}}\right\}$$

$$= \frac{z}{2}\left\{\frac{z-e^{-\theta} + z-e^\theta}{(z-e^\theta)(z-e^{-\theta})}\right\}$$

$$= \frac{z}{2} \left\{ \frac{2z - (e^{\theta} + e^{-\theta})}{z^2 - z(e^{\theta} + e^{-\theta}) + 1} \right\}$$

$$= \frac{z}{2} \left\{ \frac{2z - 2 \cosh \theta}{z^2 - 2z \cosh \theta + 1} \right\}$$

Thus,
$$Z_T(\cosh \theta) = \frac{z(z - \cosh \theta)}{z^2 - 2z \cosh \theta + 1}$$

Next,
$$\sinh \theta = \frac{e^{\theta} - e^{-\theta}}{2}$$

Proceeding on the same lines as before, we have,

$$Z_T(\sinh \theta) = \frac{z}{2} \left\{ \frac{1}{z - e^{\theta}} - \frac{1}{z - e^{-\theta}} \right\}$$

$$= \frac{z}{2} \left\{ \frac{z - e^{-\theta} - z + e^{\theta}}{z^2 - 2z \cosh \theta + 1} \right\}$$

$$= \frac{z}{2} \cdot \frac{(e^{\theta} - e^{-\theta})}{z^2 - 2z \cosh \theta + 1}$$

$$= \frac{z}{2} \cdot \frac{2 \sinh \theta}{z^2 - 2z \cosh \theta + 1}$$

Thus,
$$Z_T(\sinh \theta) = \frac{z \sinh \theta}{z^2 - 2z \cosh \theta + 1}$$

⑤ Find the Z-transform of $2n + \sin(n\pi/4) + 1$

→ Let $u_n = 2n + \sin(n\pi/4) + 1$

$$Z_T(u_n) = 2 Z_T(n) + Z_T[\sin(n\pi/4)] + Z_T(1)$$

ie.
$$Z_T(u_n) = \frac{2z}{(z-1)^2} + Z_T[\sin(n\pi/4)] + \frac{z}{z-1} \quad \text{--- ①}$$

we shall find,
$$Z_T[\sin(n\pi/4)]$$

we have $e^{in\pi/4} = \cos(n\pi/4) + i \sin(n\pi/4)$

but, $e^{in\pi/4} = (e^{i\pi/4})^n = k^n$ (say) where $k = e^{i\pi/4}$

w.k.t $Z_T(k^n) = \frac{z}{z-k}$

$$\therefore Z_T(e^{in\pi/4}) = \frac{z}{z - e^{i\pi/4}}$$

$$\text{i.e., } Z_T[\cos(n\pi/4) + i \sin(n\pi/4)] = \frac{2z}{z - \{\cos(n\pi/4) + i \sin(n\pi/4)\}}$$

$$= \frac{z}{z - \frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}}}$$

$$Z_T[\cos(n\pi/4) + i \sin(n\pi/4)] = \frac{z \left[\left(z - \frac{1}{\sqrt{2}} \right) + \frac{i}{\sqrt{2}} \right]}{\left(z - \frac{1}{\sqrt{2}} \right)^2 + \frac{1}{2}}$$

$$\therefore Z_T[\cos(n\pi/4) + i \sin(n\pi/4)] = \frac{z \left[\left(z - \frac{1}{\sqrt{2}} \right) + \frac{i}{\sqrt{2}} \right]}{z^2 - \sqrt{2}z + 1}$$

Equating the imaginary parts on both sides we get,

$$Z_T[\sin(n\pi/4)] = \frac{z/\sqrt{2}}{z^2 - \sqrt{2}z + 1} = \frac{z}{\sqrt{2}(z^2 - \sqrt{2}z + 1)}$$

we substitute this result in ①.

$$\text{Thus, } Z_T(u_n) = \frac{2z}{(z-1)^2} + \frac{z}{\sqrt{2}(z^2 - \sqrt{2}z + 1)} + \frac{z}{z-1}$$

where $u_n = 2n + \sin(n\pi/4) + 1$

⑥ find the Z-transform of $\cos(n\pi/2 + \pi/4)$

→ let $u_n = \cos(n\pi/2 + \pi/4)$

$$= \cos(n\pi/2) \cos(\pi/4) - \sin(n\pi/2) \sin(\pi/4)$$

i.e., $u_n = \frac{1}{\sqrt{2}} [\cos(n\pi/2) - \sin(n\pi/2)]$

$$\therefore Z_T(u_n) = \frac{1}{\sqrt{2}} [Z_T \cos(n\pi/2) - Z_T \sin(n\pi/2)] \quad \text{--- (1)}$$

consider, $e^{i(n\pi/2)} = (e^{i\pi/2})^n = k^n$ (say)

where $k = e^{i\pi/2}$

We know that, $Z_T(k^n) = \frac{z}{z-k}$ & hence we have,

$$Z_T(e^{in\pi/2}) = \frac{z}{z - e^{i\pi/2}} = \frac{z}{z - \cos(\pi/2) - i \sin(\pi/2)} = \frac{z}{z - i}$$

i.e., $Z_T(e^{in\pi/2}) = \frac{z(z+i)}{(z-i)(z+i)} = \frac{z^2 + iz}{z^2 + 1}$

i.e., $Z_T[\cos(n\pi/2) + i \sin(n\pi/2)] = \frac{z^2}{z^2 + 1} + i \frac{z}{z^2 + 1}$

$\Rightarrow Z_T[\cos(n\pi/2)] = \frac{z^2}{z^2 + 1}$ &

$$Z_T[\sin(n\pi/2)] = \frac{z}{z^2 + 1}$$

we substitute these results in (1).

Then,
$$Z_T(u_n) = \frac{1}{\sqrt{2}} \left[\frac{z^2}{z^2 + 1} - \frac{z}{z^2 + 1} \right] = \frac{z(z-1)}{\sqrt{2}(z^2 + 1)}$$

where $u_n = \cos(n\pi/2 + n\pi/4)$

Note:- It is advisable to remember the following result

$$Z_T[\sin(n\pi/2)] = \frac{z}{z^2+1} \quad \& \quad Z_T[\cos(n\pi/2)] = \frac{z^2}{z^2+1}$$

7) Find the Z-transform of $\sin(3n+5)$

→ Let $u_n = \sin(3n+5) = \sin 3n \cos 5 + \cos 3n \sin 5$

$$\therefore Z_T(u_n) = \cos 5 Z_T(\sin 3n) + \sin 5 Z_T(\cos 3n) \quad \text{--- ①}$$

consider, $e^{i(3n)} = (e^{3i})^n = k^n$ (say) where $k = e^{3i}$

we know that, $Z_T(k^n) = \frac{z}{z-k}$

$$\begin{aligned} \text{ie, } Z_T(e^{3in}) &= \frac{z}{z-e^{3i}} = \frac{z}{(z-\cos 3) - i\sin 3} \\ &= \frac{z[(z-\cos 3) + i\sin 3]}{(z-\cos 3)^2 + \sin^2 3} \end{aligned}$$

$$\text{ie, } Z_T(\cos 3n + i\sin 3n) = \frac{z(z-\cos 3)}{z^2-2z\cos 3+1} + i \frac{z\sin 3}{z^2-2z\cos 3+1}$$

$$\Rightarrow Z_T(\cos 3n) = \frac{z(z-\cos 3)}{z^2-2z\cos 3+1} \quad \&$$

$$Z_T(\sin 3n) = \frac{z\sin 3}{z^2-2z\cos 3+1}$$

Substituting these results in ① we get,

$$\begin{aligned} Z_T(u_n) &= \cos 5 \cdot \frac{z\sin 3}{z^2-2z\cos 3+1} + \sin 5 \cdot \frac{z(z-\cos 3)}{z^2-2z\cos 3+1} \\ &= \frac{z(\sin 3\cos 5 - \cos 3\sin 5) + z^2\sin 5}{z^2-2z\cos 3+1} \end{aligned}$$

$$= \frac{z(-\sin 2) + z^2 \sin 5}{z^2 - 2z \cos 3 + 1}$$

(18)

Thus, $Z_T(u_n) = \frac{z(z \sin 5 - \sin 2)}{z^2 - 2z \cos 3 + 1}$ where $u_n = \sin(3n\pi/5)$.

(8) Find the Z-transform of $\cosh(n\pi/2 + \theta)$

→ Let $u_n = \cosh(n\pi/2 + \theta) = \frac{1}{2} [e^{(n\pi/2 + \theta)} + e^{-(n\pi/2 + \theta)}]$

i.e., $u_n = \frac{1}{2} [e^\theta e^{n\pi/2} + e^{-\theta} e^{-n\pi/2}]$

$\therefore Z_T(u_n) = \frac{1}{2} [e^\theta Z_T(e^{n\pi/2}) + e^{-\theta} Z_T(e^{-n\pi/2})]$ — (1)

we have, $Z_T(k^n) = \frac{z}{z-k}$

Putting, $k = e^{\pi/2}$ & $e^{-\pi/2}$ we have,

$Z_T(e^{n\pi/2}) = \frac{z}{z - e^{\pi/2}}$ & $Z_T(e^{-n\pi/2}) = \frac{z}{z - e^{-\pi/2}}$

Hence (1) becomes,

$Z_T(u_n) = \frac{1}{2} \left[e^\theta \cdot \frac{z}{z - e^{\pi/2}} + e^{-\theta} \cdot \frac{z}{z - e^{-\pi/2}} \right]$

$= \frac{z}{2} \left[\frac{e^\theta (z - e^{-\pi/2}) + e^{-\theta} (z - e^{\pi/2})}{(z - e^{\pi/2})(z - e^{-\pi/2})} \right]$

$= \frac{z}{2} \left[\frac{z(e^\theta + e^{-\theta}) - \{e^{(\pi/2 - \theta)} + e^{-(\pi/2 - \theta)}\}}{z^2 - z(e^{\pi/2} + e^{-\pi/2}) + 1} \right]$

$= \frac{z}{2} \left[\frac{2z \cosh \theta - 2 \cosh(\pi/2 - \theta)}{z^2 - 2z \cosh(\pi/2) + 1} \right]$

Then,
$$Z_T(u_n) = \frac{z^2 \cosh \theta - z \cosh(\pi/2 - \theta)}{z^2 - 2z \cosh(\pi/2) + 1} //$$

where $u_n = \cosh(\pi \cdot \pi/2 + n\theta)$.

⑨ Using $Z_T(u_n) = \frac{z^2 + z}{(z-1)^3}$ & $Z_T[(n+1)^2] = \frac{z^3 + z^2}{(z-1)^3}$

→ Let $u_n = n^2$ & we have $Z_T(u_n) = Z_T(n^2) = \bar{u}(z)$

consider the property,

$$Z_T(u_{n+1}) = z [\bar{u}(z) - u_0] \quad \text{--- ①}$$

since, $u_n = n^2$, $u_{n+1} = (n+1)^2$ & $u_0 = 0$ -- hence ① becomes.

$$Z_T[(n+1)^2] = z \left[\frac{z^2 + z}{(z-1)^3} - 0 \right]$$

Then,
$$Z_T[(n+1)^2] = \frac{z^3 + z^2}{(z-1)^3} //$$

⑩ Show that $Z_T\left[\frac{1}{n!}\right] = e^{z^{-1}}$, hence find $Z_T\left[\frac{1}{(n+1)!}\right]$ &

$$Z_T\left[\frac{1}{(n+2)!}\right]$$

→ By the definition,

$$Z_T\left[\frac{1}{n!}\right] = \sum_{n=0}^{\infty} \frac{1}{n!} z^{-n} = 1 + \frac{z^{-1}}{1!} + \frac{z^{-2}}{2!} + \frac{z^{-3}}{3!} + \dots$$

but $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$ & here we have $x = z^{-1}$

Then
$$Z_T\left[\frac{1}{n!}\right] = e^{z^{-1}} = e^{1/z}$$

We have the properties,

$$Z_T[u_{n+1}] = z[\bar{u}(z) - u_0] \quad \text{--- (1)}$$

$$Z_T[u_{n+2}] = z[\bar{u}(z) - u_0 - u_1 z^{-1}] \quad \text{--- (2)}$$

Let $u_n = \frac{1}{n!}$ $\therefore Z_T(u_n) = \bar{u}(z) = e^{1/z}$

Also, $u_0 = \frac{1}{0!} = 1$ & $u_1 = \frac{1}{1!} = 1$

Then by using these results on (1) & (2) we obtain,

$$Z_T\left[\frac{1}{(n+1)!}\right] = z[e^{1/z} - 1]$$

$$Z_T\left[\frac{1}{(n+2)!}\right] = z[e^{1/z} - 1 - z^{-1}]$$

(11) Prove that $Z_T[e^{-an} u_n] = \bar{u}(e^a z)$ given that $Z_T(u_n) = \bar{u}(z)$

→ by the definition, $Z_T[e^{-an} u_n] = \sum_{n=0}^{\infty} e^{-an} u_n z^{-n} = \sum_{n=0}^{\infty} u_n (e^a)^{-n} z^{-n}$

$$Z_T[e^{-an} u_n] = \sum_{n=0}^{\infty} u_n (e^a z)^{-n} = \bar{u}(e^a z)$$

Then, $\boxed{Z_T(e^{-an} u_n) = \bar{u}(e^a z)}$

* note: (i) This result can be deduced from the damping rule by taking $v = e^a$

(ii) This result is also referred to as the exponential shifting rule.

(12) Find the z -transform of the unit step sequence
 $u(n) = \begin{cases} 0, & n < 0 \\ 1, & n \geq 0 \end{cases}$

$$\rightarrow Z_T[u(n)] = \sum_{n=0}^{\infty} u(n) z^{-n}$$

$$= \sum_{n=0}^{\infty} 1 \cdot z^{-n}$$

$$= 1 + (1/2) + (1/2)^2 + (1/2)^3 + \dots \text{ is a}$$

geometric series.

$$\therefore Z_T[u(n)] = \frac{1}{1 - (1/2)} = \frac{z}{z - 1}$$

Thus, $\boxed{Z_T[u(n)] = \frac{z}{z - 1}}$

(13) Find the z -transform of the unit impulse sequence

$$\delta(n) = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

$$\rightarrow Z_T[\delta(n)] = \sum_{n=0}^{\infty} \delta(n) z^{-n}$$

$$Z_T[\delta(n)] = \delta(0)z^0 + \delta(1)z^{-1} + \delta(2)z^{-2} + \dots$$

$$= 1 + 0 + 0 + \dots$$

$$= 1$$

Thus, $\boxed{Z_T[\delta(n)] = 1}$

(14) If $Z_T(u_n) = \bar{u}(z)$, show that $Z_T(nu_n) = -z \frac{d}{dz} [\bar{u}(z)]$ (22)

$$\begin{aligned} \rightarrow Z_T(nu_n) &= \sum_{n=0}^{\infty} (nu_n) z^{-n} \\ &= -z \sum_{n=0}^{\infty} -nu_n z^{-n-1} \end{aligned}$$

$$\begin{aligned} Z_T(nu_n) &= -z \sum_{n=0}^{\infty} u_n \frac{d}{dz} (z^{-n}) \\ &= -z \sum_{n=0}^{\infty} \frac{d}{dz} (u_n z^{-n}) \\ &= -z \frac{d}{dz} \sum_{n=0}^{\infty} u_n z^{-n} \\ &= -z \frac{d}{dz} [\bar{u}(z)] \end{aligned}$$

$$\text{Thus, } Z_T(nu_n) = -z \frac{d}{dz} [\bar{u}(z)] = -z \frac{d}{dz} [Z_T(u_n)]$$

$$\boxed{Z_T(nu_n) = -z \frac{d}{dz} [Z_T(u_n)]}$$

* Remark: In general we can show that

$$Z_T[n^r u_n] = (-1)^r z^r \frac{d^r}{dz^r} [\bar{u}(z)] \text{ where } r \text{ is a +ve integer.}$$

(15) Find $Z_T\left[\frac{1}{n+1}\right]$

$$\begin{aligned} \rightarrow Z_T\left[\frac{1}{n+1}\right] &= \sum_{n=0}^{\infty} \frac{1}{n+1} z^{-n} \\ &= 1 + \frac{1}{2} z^{-1} + \frac{1}{3} z^{-2} + \frac{1}{4} z^{-3} + \dots \\ &= 1 + \frac{1}{2z} + \frac{1}{3z^2} + \frac{1}{4z^3} + \dots \end{aligned}$$

$$Z_T \left[\frac{1}{n+1} \right] = z \left[\frac{1}{z} + \frac{1}{2z^2} + \frac{1}{3z^3} + \frac{1}{4z^4} + \dots \right] \quad \text{--- (1)}$$

we have, $\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$

$$\log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \dots$$

$$-\log(1-x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \dots$$

By taking $x = \frac{1}{z}$ we have,

$$-\log \left[1 - \frac{1}{z} \right] = \frac{1}{z} + \frac{1}{2z^2} + \frac{1}{3z^3} + \frac{1}{4z^4} + \dots$$

$$-\log \left[\frac{z-1}{z} \right] = \frac{1}{z} + \frac{1}{2z^2} + \frac{1}{3z^3} + \frac{1}{4z^4} + \dots$$

$$\log \left[\frac{z}{z-1} \right] = \frac{1}{z} + \frac{1}{2z^2} + \frac{1}{3z^3} + \frac{1}{4z^4} + \dots \rightarrow \text{(2)}$$

we use (2) in RHS of (1)

$$\text{Thus, } \boxed{Z_T \left[\frac{1}{n+1} \right] = z \log \left[\frac{z}{z-1} \right]} //$$

(16) Find $Z_T \left[\frac{1}{(n+1)(n+2)} \right]$

$\rightarrow$ Let $u_n = \frac{1}{(n+1)(n+2)}$

$$\frac{1}{(n+1)(n+2)} = \frac{1}{(n+1)} - \frac{1}{(n+2)}, \text{ by partial fractions.}$$

$$\text{now, } Z_T[u_n] = Z_T \left[\frac{1}{n+1} \right] - Z_T \left[\frac{1}{n+2} \right] \quad \text{--- (3)}$$

but, $Z_T \left[\frac{1}{n+1} \right] = z \log \left[\frac{z}{z-1} \right] \quad \text{--- (2)}$

or $Z_T \left[\frac{1}{n+2} \right] = \sum_{n=0}^{\infty} \frac{1}{n+2} z^{-n}$
 $= \frac{1}{2} + \frac{z^{-1}}{3} + \frac{z^{-2}}{4} + \dots$

i.e. $Z_T \left[\frac{1}{n+2} \right] = \frac{1}{2} + \frac{x}{3} + \frac{x^2}{4} + \dots$ where $x = z^{-1} = \frac{1}{z}$

$Z_T \left[\frac{1}{n+2} \right] = \frac{1}{x^2} \left[\frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \dots \right]$

$Z_T \left[\frac{1}{n+2} \right] = \frac{1}{x^2} \{ -\log(1-x) - x \}$
 $= -z^2 \log \left(1 - \frac{1}{z} \right) - z$

$\therefore Z_T \left[\frac{1}{n+2} \right] = -z^2 \log \left[\frac{z-1}{z} \right] - z \quad \text{--- (3)}$

using (2) & (3) for (1) we have,

$Z_T(u_n) = z \log \left[\frac{z}{z-1} \right] + z^2 \log \left[\frac{z-1}{z} \right] + z$

i.e., $Z_T(u_n) = z \log \left[\frac{z}{z-1} \right] - z^2 \log \left[\frac{z}{z-1} \right] + z$

Thus, $\boxed{Z_T \left[\frac{1}{(n+1)(n+2)} \right] = z \log \left[\frac{z}{z-1} \right] \{ 1-z \} + z}$ //

* INITIAL VALUE THEOREM

Statement: If $Z_T(u_n) = \bar{u}(z)$ then $\lim_{z \rightarrow \infty} \bar{u}(z) = u_0$

proof: we have by the definition,

$$Z_T(u_n) = \sum_{n=0}^{\infty} u_n z^{-n}$$

$$\text{i.e., } \bar{u}(z) = u_0 + u_1 z^{-1} + u_2 z^{-2} + u_3 z^{-3} + \dots \quad \text{--- (1)}$$

$$\begin{aligned} \therefore \lim_{z \rightarrow \infty} \bar{u}(z) &= \lim_{z \rightarrow \infty} \left[u_0 + \frac{u_1}{z} + \frac{u_2}{z^2} + \frac{u_3}{z^3} + \dots \right] \\ &= u_0 + 0 + 0 + \dots \\ &= u_0 \end{aligned}$$

$$\text{Then, } \lim_{z \rightarrow \infty} \bar{u}(z) = u_0$$

Why we can also obtain other initial values $u_1, u_2, \dots$ as follows.

we have from (1),

$$\bar{u}(z) - u_0 = \frac{u_1}{z} + \frac{u_2}{z^2} + \frac{u_3}{z^3} + \dots$$

$$z [\bar{u}(z) - u_0] = u_1 + \frac{u_2}{z} + \frac{u_3}{z^2} + \dots$$

$$\therefore \lim_{z \rightarrow \infty} z [\bar{u}(z) - u_0] = u_1$$

$$\text{Also, } \bar{u}(z) - u_0 - \frac{u_1}{z} = \frac{u_2}{z^2} + \frac{u_3}{z^3} + \dots$$

$$\therefore \lim_{z \rightarrow \infty} z^2 \left[\bar{u}(z) - u_0 - \frac{u_1}{z} \right] = u_2$$

* FINAL VALUE THEOREM

(26)

Statement: If $Z_T(u_n) = \bar{u}(z)$ then, $\lim_{z \rightarrow 1} [(z-1)\bar{u}(z)] = \lim_{n \rightarrow \infty} u_n$

Proof: we have the result,

$$Z_T[u_{n+1}] = z[\bar{u}(z) - u_0] \quad \text{--- (1)}$$

$$\text{Also, } Z_T(u_n) = \bar{u}(z) \quad \text{--- (2)}$$

now, (1) - (2) will give us,

$$Z_T[u_{n+1}] - Z_T(u_n) = z\bar{u}(z) - zu_0 - \bar{u}(z)$$

$$(1) \quad Z_T[u_{n+1} - u_n] = (z-1)\bar{u}(z) - zu_0$$

$$\text{i.e., } \sum_{n=0}^{\infty} [u_{n+1} - u_n] z^{-n} = (z-1)\bar{u}(z) - zu_0$$

now taking limit as $z \rightarrow 1$ we have,

$$\lim_{z \rightarrow 1} \sum_{n=0}^{\infty} (u_{n+1} - u_n) \frac{1}{z^n} = \lim_{z \rightarrow 1} [(z-1)\bar{u}(z) - zu_0]$$

$$\text{i.e., } \sum_{n=0}^{\infty} (u_{n+1} - u_n) = \lim_{z \rightarrow 1} [(z-1)\bar{u}(z)] - u_0$$

$$\lim_{z \rightarrow 1} [(z-1)\bar{u}(z)] = u_0 + \sum_{n=0}^{\infty} (u_{n+1} - u_n)$$

$$\lim_{z \rightarrow 1} [(z-1)\bar{u}(z)] = u_0 + \lim_{m \rightarrow \infty} \sum_{n=0}^m (u_{n+1} - u_n)$$

$$= u_0 + \lim_{m \rightarrow \infty} \{ (u_1 - u_0) + (u_2 - u_1) + (u_3 - u_2) + \dots + (u_m - u_{m-1}) + (u_{m+1} - u_m) \}$$

$$= u_0 + \lim_{m \rightarrow \infty} (-u_0 + u_{m+1})$$

$$= \lim_{m \rightarrow \infty} u_{m+1}$$

It should be noted that as $m \rightarrow \infty$, $(m+1)$ also tends to ∞ & hence this is equivalent to $\lim_{n \rightarrow \infty} u_n$ (27)

Thus we have proved that,

$$\lim_{z \rightarrow 1} [(z-1)\bar{u}(z)] = \lim_{n \rightarrow \infty} u_n$$

* PROBLEMS

① If $\bar{u}(z) = \frac{2z^2 + 3z + 12}{(z-1)^4}$, find the value of u_0, u_1, u_2, u_3 .

→ We have by the definition of Z-transforms,

$$\bar{u}(z) = u_0 + u_1 z^{-1} + u_2 z^{-2} + u_3 z^{-3} + u_4 z^{-4} + \dots$$

$$\text{i.e. } \bar{u}(z) = u_0 + \frac{u_1}{z} + \frac{u_2}{z^2} + \frac{u_3}{z^3} + \frac{u_4}{z^4} + \dots$$

therefore we have,

$$u_0 = \lim_{z \rightarrow \infty} \bar{u}(z) \quad \text{--- (1)}$$

$$u_1 = \lim_{z \rightarrow \infty} z [\bar{u}(z) - u_0] \quad \text{--- (2)}$$

$$u_2 = \lim_{z \rightarrow \infty} z^2 [\bar{u}(z) - u_0 - \frac{u_1}{z}] \quad \text{--- (3)}$$

$$u_3 = \lim_{z \rightarrow \infty} z^3 [\bar{u}(z) - u_0 - \frac{u_1}{z} - \frac{u_2}{z^2}] \quad \text{--- (4)}$$

$$\begin{aligned} \text{from (1), } u_0 &= \lim_{z \rightarrow \infty} \frac{2z^2 + 3z + 12}{(z-1)^4} \\ &= \lim_{z \rightarrow \infty} \frac{z^2 (2 + 3/z + 12/z^2)}{z^4 (1 - 1/z)^4} \end{aligned}$$

$$= \lim_{z \rightarrow \infty} \frac{1}{z^2} \cdot \frac{(2 + 3/z + 12/z^2)}{(1 - 1/z)^4}$$

$$= 0 \cdot \frac{2}{1} = 0$$

$$\therefore \boxed{u_0 = 0}$$

From (2), $u_1 = \lim_{z \rightarrow \infty} z \cdot \frac{3z^2 + 3z + 12}{(z-1)^4}$, since $u_0 = 0$

$$= \lim_{z \rightarrow \infty} z^3 \cdot \frac{(2 + 3/z + 12/z^2)}{z^4 (1 - 1/z)^4}$$

$$= \lim_{z \rightarrow \infty} \frac{1}{z} \cdot \frac{(2 + 3/z + 12/z^2)}{(1 - 1/z)^4} = 0$$

$$\therefore \boxed{u_1 = 0}$$

From (3), $u_2 = \lim_{z \rightarrow \infty} z^2 \left[\frac{3z^2 + 3z + 12}{(z-1)^4} \right]$ since $u_0 = 0 = u_1$

$$= \lim_{z \rightarrow \infty} z^4 \cdot \frac{(2 + 3/z + 12/z^2)}{z^4 (1 - 1/z)^4}$$

$$= \frac{2 + 0 + 0}{(1 - 0)^4}$$

$$= 2$$

$$\therefore \boxed{u_2 = 2}$$

From (4), $u_3 = \lim_{z \rightarrow \infty} z^3 \left[\frac{3z^2 + 3z + 12}{(z-1)^4} - \frac{2}{z^2} \right]$

$$u_3 = \lim_{z \rightarrow \infty} z^3 \left[\frac{3z^4 + 3z^3 + 12z^2 - 2(z^4 - 4z^3 + 6z^2 - 4z + 1)}{z^2 (z-1)^4} \right]$$

$$u_3 = \lim_{z \rightarrow \infty} z \left[\frac{11z^3 + 8z - 2}{(z-1)^4} \right]$$

$$u_3 = \lim_{z \rightarrow \infty} z^4 \cdot \frac{(11 + 8/z^2 - 2/z^3)}{z^4 (1 - 1/z)^4}$$

$$\therefore \boxed{u_3 = 11} //$$

Thus

$$\boxed{u_0 = 0, u_1 = 0, u_2 = 2 \text{ \& } u_3 = 11} //$$

⑤ ^{M.P.} Given $z_T(u_n) = \frac{2z^2 + 3z + 4}{(z-3)^3}$, $|z| > 1$ Show that

$$u_1 = 2, u_2 = 21, u_3 = 139$$

→

$$\begin{aligned} u_0 &= \lim_{z \rightarrow \infty} \frac{2z^2 + 3z + 4}{(z-3)^3} \\ &= \lim_{z \rightarrow \infty} \frac{z^2(2 + 3/z + 4/z^2)}{z^3(1 - 3/z)^3} \\ &= \lim_{z \rightarrow \infty} \frac{1}{z} \cdot \frac{(2 + 3/z + 4/z^2)}{(1 - 3/z)^3} = 0 \end{aligned}$$

$$\therefore u_0 = 0$$

$$\begin{aligned} u_1 &= \lim_{z \rightarrow \infty} z \left[\frac{2z^2 + 3z + 4}{(z-3)^3} \right] \text{ since } u_0 = 0. \\ &= \lim_{z \rightarrow \infty} z^3 \cdot \frac{(2 + 3/z + 4/z^2)}{z^3(1 - 3/z)^3} = 2 \end{aligned}$$

$$\therefore \boxed{u_1 = 2}$$

$$\begin{aligned} u_2 &= \lim_{z \rightarrow \infty} z^2 \left[\frac{2z^2 + 3z + 4}{(z-3)^3} - 0 - \frac{2}{z} \right] \\ &= \lim_{z \rightarrow \infty} z^2 \left[\frac{2z^3 + 3z^2 + 4z - 2(z^3 - 9z^2 + 27z - 27)}{z(z-3)^3} \right] \end{aligned}$$

$$u_2 = \lim_{z \rightarrow \infty} \frac{z(21z^2 - 50z + 54)}{(z-3)^3}$$

$$= \lim_{z \rightarrow \infty} \frac{z^3(21 - 50/z + 54/z^2)}{z^3(1 - 3/z)^3} = 21$$

$$\therefore \boxed{u_2 = 21}$$

$$u_3 = \lim_{z \rightarrow \infty} z^3 \left[\frac{21z^2 + 3z + 4}{(z-3)^3} - \frac{2}{z} - \frac{21}{z^2} \right]$$

$$= \lim_{z \rightarrow \infty} z^3 \left[\frac{z^3(21z^2 + 3z + 4) - 2z^2(z-3)^3 - 21z(z-3)^3}{z^3(z-3)^3} \right]$$

$$= \lim_{z \rightarrow \infty} \left[\frac{(21z^5 + 3z^4 + 4z^3) - (2z^2 + 21z)(z^3 - 9z^2 + 27z - 27)}{(z-3)^3} \right]$$

$$= \lim_{z \rightarrow \infty} \left[\frac{(21z^5 + 3z^4 + 4z^3) - (2z^5 + 3z^4 - 135z^3 + 513z^2 - 567z)}{(z-3)^3} \right]$$

$$= \lim_{z \rightarrow \infty} \left[\frac{139z^3 - 513z^2 + 567z}{(z-3)^3} \right]$$

$$= \lim_{z \rightarrow \infty} \frac{z^3(139 - 513/z + 567/z^2)}{z^3(1 - 3/z)^3} = 139$$

$$\therefore \boxed{u_3 = 139}$$

Thus we have proved that

$$\boxed{u_1 = 2, u_2 = 21 \quad \& \quad u_3 = 139}$$

③ Given $Z_T(u_n) = \frac{z}{z-1} + \frac{z}{z^2+1}$ obtain the Z-transform of u_{n+2}

→ we have, $Z_T(u_{n+2}) = z^2 \left[\bar{u}(z) - u_0 - \frac{u_1}{z} \right] \quad \text{--- ①}$

we shall first compute u_0 & u_1

$$u_0 = \lim_{z \rightarrow \infty} \bar{u}(z)$$

$$u_0 = \lim_{z \rightarrow \infty} \left[\frac{z}{z-1} + \frac{z}{z^2+1} \right]$$

$$= \lim_{z \rightarrow \infty} \left[\frac{z}{z(1-1/z)} + \frac{z}{z^2(1+1/z^2)} \right]$$

$$= \lim_{z \rightarrow \infty} \left[\frac{1}{1-1/z} + \frac{1}{z} \cdot \frac{1}{(1+1/z^2)} \right]$$

$$= 1 + 0 = 1$$

$$\therefore \boxed{u_0 = 1}$$

$$u_1 = \lim_{z \rightarrow \infty} z [\bar{u}(z) - u_0]$$

$$= \lim_{z \rightarrow \infty} z \left[\frac{z}{z-1} + \frac{z}{z^2+1} - 1 \right]$$

$$= \lim_{z \rightarrow \infty} z \left[\frac{z^3 + z + z^2 - z - (z^3 - z^2 + z - 1)}{(z-1)(z^2+1)} \right]$$

$$= \lim_{z \rightarrow \infty} z \left[\frac{2z^2 - z + 1}{(z-1)(z^2+1)} \right]$$

$$= \lim_{z \rightarrow \infty} \frac{z^3 (2 - 1/z + 1/z^2)}{z(1-1/z)z^2(1+1/z^2)} = 2$$

$$3/4_1 = 2$$

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now, from ① we have,

$$Z_T(u_{n+2}) = z^2 \left[\frac{z}{z-1} + \frac{z}{z^2+1} - 1 - \frac{2}{z} \right]$$

$$= z^2 \left[\frac{z^3 + z^2}{(z-1)(z^2+1)} - \frac{(z+2)}{z} \right]$$

$$= \frac{z^2(z^2 - z + 2)}{(z-1)(z^2+1)(z)} \quad \text{on simplification,}$$

$$\text{Thus, } \boxed{Z_T(u_{n+2}) = \frac{z(z^2 - z + 2)}{(z-1)(z^2+1)}} \quad \text{Q.E.D.}$$

* INVERSE Z-Transform

We have already stated that, if $Z_T(u_n) = \bar{u}(z)$ then $Z_T^{-1}[\bar{u}(z)] = u_n$ is called the inverse-z transform of $\bar{u}(z)$.

* Standard Inverse Z-transforms

$$(1) \quad Z_T^{-1} \left[\frac{z}{z-1} \right] = 1$$

$$(6) \quad Z_T^{-1} \left[\frac{kz^2 + k^2z}{(z-k)^3} \right] = k^n \cdot n^2$$

$$(2) \quad Z_T^{-1} \left[\frac{z}{z-k} \right] = k^n$$

$$(7) \quad Z_T^{-1} \left[\frac{z^3 + 4z^2 + z}{(z-1)^4} \right] = n^3$$

$$(3) \quad Z_T^{-1} \left[\frac{z}{(z-1)^2} \right] = n$$

$$(8) \quad Z_T^{-1} \left[\frac{kz^3 + 4k^2z^2 + k^3z}{(z-k)^4} \right] = k^n \cdot n^3$$

$$(4) \quad Z_T^{-1} \left[\frac{kz}{(z-k)^2} \right] = k^n \cdot n$$

$$(9) \quad Z_T^{-1} \left[\frac{z}{z^2+1} \right] = \sin(n\pi/2)$$

$$(5) \quad Z_T^{-1} \left[\frac{z^2+z}{(z-1)^3} \right] = n^2$$

$$(10) \quad Z_T^{-1} \left[\frac{z^2}{z^2+1} \right] = \cos(n\pi/2)$$

* Type - (1)

Inverse z-transform of rational algebraic functions by partial fractions method.

→ working procedure for problems

Step 1: Given $\bar{u}(z) = \frac{f(z)}{g(z)}$ we need to express $g(z)$ in terms of non repeated linear factors only.

Step 2: we consider $\frac{\bar{u}(z)}{z}$ in the form of a proper fraction and resolve into partial fractions.

Step 3: we multiply by z to have $\bar{u}(z)$ involving various terms of the form $c \cdot (z/(z-k))$, c being a constant.

Step 4: Finally we compute the inverse z -transform of these terms resulting in the required $z_T^{-1}[\bar{u}(z)]$.

* Note: If $g(z)$ involves repeated linear factors of the form: $(z-k)^2, (z-k)^3, (z-k)^4, \dots$ we need to take it into account the corresponding terms in the numerator: $kz, (kz^2+k^2z), (kz^3+k^2z^2+k^3z), \dots$ [by referring into inverse z -transform] respectively & express $u(z)$ suitably with terms multiplied by $A, B, C, \dots$.

we compute $A, B, C, \dots$ & find $z_T^{-1}[\bar{u}(z)]$

* Problems.

(1) Find the inverse z -transform of $\frac{z}{(z-1)(z-2)}$

$$\rightarrow \text{Let } \bar{u}(z) = \frac{z}{(z-1)(z-2)}$$

$$\therefore \frac{\bar{u}(z)}{z} = \frac{1}{(z-1)(z-2)}$$

$$\text{Let } \frac{1}{(z-1)(z-2)} = \frac{A}{z-1} + \frac{B}{z-2}$$

$$1 = A(z-2) + B(z-1)$$

$$\text{Put } z=1 \Rightarrow 1 = A(-1) \Rightarrow \boxed{A = -1}$$

put $z = 0 \Rightarrow 1 = B(1) \Rightarrow \boxed{B=1}$

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hence, $\frac{\bar{u}(z)}{z} = \frac{-1}{z-1} + \frac{1}{z-2}$

$$\bar{u}(z) = \frac{-z}{z-1} + \frac{z}{z-2}$$

$$\Rightarrow z_T^{-1}[\bar{u}(z)] = z_T^{-1}\left[\frac{z}{z-2}\right] - z_T^{-1}\left[\frac{z}{z-1}\right]$$

we know, $z_T^{-1}\left[\frac{z}{z-p}\right] = p^n$

thus, $z_T^{-1}[\bar{u}(z)] = u_n = 2^n - 1$

② compute the inverse z-transform of $\frac{3z^2+2z}{(5z-1)(5z+2)}$

→ let, $\bar{u}(z) = \frac{3z^2+2z}{(5z-1)(5z+2)}$

$$\frac{\bar{u}(z)}{z} = \frac{3z+2}{(5z-1)(5z+2)}$$

let $\frac{3z+2}{(5z-1)(5z+2)} = \frac{A}{5z-1} + \frac{B}{5z+2}$

$$3z+2 = A(5z+2) + B(5z-1)$$

put $z = 1/5 \Rightarrow 13/5 = A(3)$

$$\Rightarrow \boxed{A = 13/15}$$

put $z = -2/5 \Rightarrow 4/5 = B(-3)$

$$\boxed{B = -4/15}$$

Hence,

$$\frac{\bar{u}(z)}{z} = \frac{13}{15} \cdot \frac{1}{5z-1} - \frac{4}{75} \cdot \frac{z}{5z+2/5}$$

Thus,

$$\boxed{Z^{-1} [\bar{u}(z)] = u_n = \frac{1}{75} \{ 13(1/5)^n - 4(-2/5)^n \}}$$

Ex. 1.1.1

③ Obtain the inverse Z-transform of $\frac{9z^2+3z}{(z+2)(z-4)}$

→ Let $\bar{u}(z) = \frac{9z^2+3z}{(z+2)(z-4)}$

$$\therefore \frac{\bar{u}(z)}{z} = \frac{9z+3}{(z+2)(z-4)}$$

$$\text{Let } \frac{9z+3}{(z+2)(z-4)} = \frac{A}{(z+2)} + \frac{B}{z-4}$$

$$9z+3 = A(z-4) + B(z+2)$$

$$\text{put } z = -2 \Rightarrow -1 = A(-2-4) + B(-2+2)$$

$$\Rightarrow -1 = A(-6)$$

$$\boxed{A = 1/6}$$

$$\text{put } z = 4 \Rightarrow 11 = B(6)$$

$$\boxed{B = 11/6}$$

$$\text{Hence } \frac{\bar{u}(z)}{z} = \frac{1}{6} \cdot \frac{1}{z+2} + \frac{11}{6} \cdot \frac{1}{z-4}$$

$$\bar{u}(z) = \frac{1}{6} \cdot \frac{z}{z+2} + \frac{11}{6} \cdot \frac{z}{z-4}$$

$$\Rightarrow Z_T^{-1} [\bar{u}(z)] = \frac{1}{6} Z_T^{-1} \left[\frac{z}{z+2} \right] + \frac{11}{6} Z_T^{-1} \left[\frac{z}{z-4} \right]$$

$$\text{Thus, } \boxed{Z_T^{-1} [\bar{u}(z)] = u_n = \frac{1}{6} \{ (-2)^n + 11(4)^n \}}$$

✓ up

④ find $Z_T^{-1} \left[\frac{8z - z^3}{(4-z)^3} \right]$

→ Here the denominator has repeated linear factors

$$\text{Let } \bar{u}(z) = \frac{8z - z^3}{(4-z)^3} = \frac{z^3 - 8z}{(z-4)^3}$$

$$\text{we have, } Z_T^{-1} \left[\frac{z}{z-4} \right] = 4^n,$$

$$Z_T^{-1} \left[\frac{4z}{(z-4)^2} \right] = 4^n \cdot n$$

$$\& Z_T^{-1} \left[\frac{4z^2 + 16z}{(z-4)^3} \right] = 4^n \cdot n^2$$

we resolve $\bar{u}(z)$ as follows,

$$\text{Let, } \frac{z^3 - 8z}{(z-4)^3} = \frac{A z}{(z-4)} + \frac{B \cdot 4z}{(z-4)^2} + \frac{C \cdot 4z^2 + 16z}{(z-4)^3} \quad \text{--- ①}$$

$$\frac{z^3 - 8z}{(z-4)^3} = \frac{A z (z-4)^2 + 4B z (z-4) + 4C z (z+4)}{(z-4)^3}$$

$$z^2 - 8 = A(z-4)^2 + 4B(z-4) + 4C(z+4)$$

$$\text{Put } z=4 \Rightarrow 8 = 4C(8) \Rightarrow \boxed{C = 1/4}$$

By equating the coefficient of z^2 on both side we have,

$$A = 1.$$

Also by equating the coefficient of z on both sides we have,

$$-8A + 4B + 4C = 0$$

$$-8 + 4B + 4 = 0$$

$$\therefore \boxed{B = 7/4}$$

$\therefore$ Substituting the values of A, B in (1) we have,

$$\frac{z^3 - 8z}{(z-4)^3} = \frac{z}{z-4} + \frac{7}{4} \cdot \frac{4z}{(z-4)^2} + \frac{1}{4} \cdot \frac{4z^2 + 16z}{(z-4)^3}$$

$$\Rightarrow Z_T^{-1} \left[\frac{z^3 - 8z}{(z-4)^3} \right] = Z_T^{-1} \left[\frac{z}{z-4} \right] + \frac{7}{4} Z_T^{-1} \left[\frac{4z}{(z-4)^2} \right] + \frac{1}{4} Z_T^{-1} \left[\frac{4z^2 + 16z}{(z-4)^3} \right]$$

$$\text{i.e., } Z_T^{-1} [\bar{u}(z)] = u_n = 4^n + \frac{7}{4} 4^n \cdot n + \frac{1}{4} 4^n \cdot n^2$$

$$\text{Thus, } Z_T^{-1} [\bar{u}(z)] = u_n = \frac{4^n}{4} [4 + 7n + n^2]$$

$$\boxed{Z_T^{-1} [\bar{u}(z)] = u_n = 4^{n-1} [4 + 7n + n^2]}$$

(5) Find the inverse z -transform of $\frac{z^3 - 20z}{(z-2)^3(z-4)}$

$$\rightarrow \text{Let } \bar{u}(z) = \frac{z^3 - 20z}{(z-2)^3(z-4)}$$

$$\text{we have, } Z_T^{-1} \left[\frac{z}{z-2} \right] = 2^n, \quad Z_T^{-1} \left[\frac{2z}{(z-2)^2} \right] = 2^n \cdot n$$

$$Z_T^{-1} \left[\frac{2z^2 + 4z}{(z-2)^3} \right] = 2^n \cdot n^2, \quad Z_T^{-1} \left[\frac{z}{z-4} \right] = 4^n$$

we resolve $\bar{u}(z)$ as follows.

$$\bar{u}(z) = \frac{z^3 - 20z}{(z-2)^3(z-4)} = A \cdot \frac{z}{z-2} + B \frac{2z}{(z-2)^2} + C \frac{2z^2+4z}{(z-2)^3} + D \cdot \frac{z}{(z-4)} \quad \text{--- (1)}$$

$$\frac{z^3 - 20z}{(z-2)^3(z-4)} = \frac{Az(z-2)^2(z-4) + 2Bz(z-2)(z-4) + C(2z^2+4z)(z-4) + Dz(z-2)^3}{(z-2)^3(z-4)}$$

$$z^3 - 20z = A(z-2)^2(z-4) + 2B(z-2)(z-4) + C(2z^2+4z)(z-4) + D(z-2)^3$$

put $z=2 \Rightarrow -16 = -16C$

$$\Rightarrow \boxed{C=1}$$

put $z=4 \Rightarrow -4 = D(8) \Rightarrow \boxed{D=-\frac{1}{2}}$

Equating the coefficient of z^3 on both side we have,

$$A + D = 0 \quad \therefore A = \frac{1}{2}$$

put $z=0 \Rightarrow -20 = A(4)(-4) + 2B(8) + C(-16) + D(-8)$

$$-20 = -8 + 16B - 16 + 4$$

$$\therefore \boxed{B=0}$$

substituting the values of A, B, C, D in (1) & taking inverse we have,

$$\begin{aligned} z_T^{-1}[\bar{u}(z)] &= \frac{1}{2} z_T^{-1} \left[\frac{z}{z-2} \right] + z_T^{-1} \left[\frac{2z^2+4z}{(z-2)^3} \right] - \frac{1}{2} z_T^{-1} \left[\frac{z}{z-4} \right] \\ &= \frac{1}{2} \cdot 2^n + 2^n \cdot n^2 - \frac{1}{2} 4^n \end{aligned}$$

Thus, $z_T^{-1}[\bar{u}(z)] = u_n = 2^{n-1} + 2^n \cdot n^2 - 2^{n-1}$

* Type - (2) : power series method.

working procedure for problems

Step ①: The given $\bar{u}(z)$ is modified into a suitable form so as to expand it as an infinite series by using some of the standard expansions like

$$(i) (1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots$$

$$(ii) \log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

$$(iii) e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

Step ②: $\bar{u}(z)$ is put in the form $\sum_{n=0}^{\infty} u_n z^{-n}$

$$\text{that is } \bar{u}(z) = Z_T(u_n)$$

Step ③: The required $Z_T^{-1}[\bar{u}(z)]$ will be u_n .

* Problems

① Find the inverse z -transform of

$$(a) \log\left[\frac{z}{z+1}\right] \quad (b) z \log\left[\frac{z}{z+1}\right]$$

$$\rightarrow (a) \text{ let } \bar{u}(z) = \log\left[\frac{z}{z+1}\right]$$

$$\bar{u}(z) = \log\left[\frac{z}{z(1+\frac{1}{z})}\right] = -\log\left[1+\frac{1}{z}\right]$$

$$\text{we have, } \log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

$$\therefore \bar{u}(z) = - \left\{ \left(\frac{1}{z} \right) - \frac{1}{2} \left[\frac{1}{z^2} \right] + \frac{1}{3} \left[\frac{1}{z^3} \right] - \frac{1}{4} \left[\frac{1}{z^4} \right] + \dots \right\} \quad (41)$$

$$\bar{u}(z) = -\frac{1}{z} + \frac{1}{2} \left[\frac{1}{z^2} \right] - \frac{1}{3} \left[\frac{1}{z^3} \right] + \frac{1}{4} \left[\frac{1}{z^4} \right] - \dots$$

$$\bar{u}(z) = \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \frac{1}{z^n}$$

$$= \sum_{n=1}^{\infty} \frac{(-1)^n}{n} z^{-n} = \sum_{n=0}^{\infty} u_n z^{-n}$$

$$\text{Then, } u_n = Z_T^{-1} [\bar{u}(z)] = \begin{cases} \frac{(-1)^n}{n} & \text{if } n \geq 1 \\ 0 & \text{if } n = 0 \end{cases}$$

$$(b) \text{ Let, } \bar{u}(z) = z \log \left[\frac{z}{z+1} \right]$$

$$\bar{u}(z) = z \left\{ -\frac{1}{z} + \frac{1}{2} \left[\frac{1}{z^2} \right] - \frac{1}{3} \left[\frac{1}{z^3} \right] + \frac{1}{4} \left[\frac{1}{z^4} \right] - \dots \right\} \quad \text{as in (a)}$$

$$= -1 + \frac{1}{2} \left[\frac{1}{z} \right] - \frac{1}{3} \left[\frac{1}{z^2} \right] + \frac{1}{4} \left[\frac{1}{z^3} \right] - \dots$$

$$\bar{u}(z) = \sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{n+1} z^{-n}$$

$$\bar{u}(z) = \sum_{n=0}^{\infty} u_n z^{-n}$$

$$\text{Then, } \boxed{u_n = Z_T^{-1} [\bar{u}(z)] = \frac{(-1)^{n+1}}{n+1}}$$

② Obtain the inverse Z-transform of $\frac{z}{(z+1)^2}$ by power series expansion.

→ Let, $\bar{u}(z) = \frac{z}{(z+1)^2}$

$$\bar{u}(z) = \frac{z}{z^2 \left[1 + \frac{1}{z}\right]^2} = \frac{1}{z} \left[1 + \frac{1}{z}\right]^{-2}$$

we have, $(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots$

$$\therefore (1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \dots$$

now, $\bar{u}(z) = \frac{1}{z} \left\{ 1 - 2 \cdot \frac{1}{z} + 3 \cdot \frac{1}{z^2} - 4 \cdot \frac{1}{z^3} + \dots \right\}$

$$\bar{u}(z) = \frac{1}{z} - 2 \frac{1}{z^2} + 3 \frac{1}{z^3} - 4 \frac{1}{z^4} + \dots$$

i.e., $\bar{u}(z) = \sum_{n=0}^{\infty} (-1)^{n-1} n z^{-n}$

$$= \sum_{n=0}^{\infty} u_n z^{-n}$$

Then,
$$u_n = Z_T^{-1}[\bar{u}(z)] = (-1)^{n-1} \cdot n$$

③ Prove the following

① $Z_T^{-1}[e^{yz}] = \frac{1}{n!}$ ② $Z_T^{-1}[z(e^{yz} - 1)] = \frac{1}{(n+1)!}$

→ ① $e^{yz} = 1 + \frac{1}{1!} \left[\frac{1}{z} \right] + \frac{1}{2!} \frac{1}{z^2} + \frac{1}{3!} \frac{1}{z^3} + \dots$

$$\text{i.e., } e^{1/2} = \sum_{n=0}^{\infty} \frac{1}{n!} z^{-n} = Z_T \left[\frac{1}{n!} \right]$$

$$\text{Thus, } Z_T^{-1} [e^{1/2}] = \frac{1}{n!}$$

$$\textcircled{b} \quad Z_T^{-1} [z(e^{1/2} - 1)] = \frac{1}{(n+1)!}$$

$$\rightarrow z(e^{1/2} - 1) = z \left[\frac{1}{1!} \cdot \frac{1}{z} + \frac{1}{2!} \frac{1}{z^2} + \frac{1}{3!} \frac{1}{z^3} + \dots \right]$$

$$z(e^{1/2} - 1) = \frac{1}{1!} + \frac{1}{2!} \frac{1}{z} + \frac{1}{3!} \frac{1}{z^2} + \dots$$

$$\text{i.e., } z(e^{1/2} - 1) = \sum_{n=0}^{\infty} \frac{1}{(n+1)!} z^{-n}$$

$$= Z_T \left[\frac{1}{(n+1)!} \right]$$

$$\text{Thus, } Z_T^{-1} [z(e^{1/2} - 1)] = \frac{1}{(n+1)!}$$

* Application of Z-transforms to solve difference equations.

→ Working Procedure for problems

Step ①: we take Z-transforms on both sides of the given difference equation.

Step ②: we use the known expressions for the Z-transform for the terms like u_{n+2} , u_{n+1} .

Step ③: we obtain $\bar{u}(z) = Z_T(u_n)$ as a function of z .

Step ④: The required solution $u_n = Z_T^{-1}[\bar{u}(z)]$.

* note:

→ If the initial values $u_0, u_1, \dots$ are not given we get the general solution of the given difference equation.

→ If the values $u_0, u_1, \dots$ are given specifically, we use them in the expressions of Z-transforms of u_{n+2} , u_{n+1} & obtain $\bar{u}(z) = Z_T(u_n)$.

Further, the solution $u_n = Z_T^{-1}[\bar{u}(z)]$ so obtained will be the particular solution of the given difference equation.

* PROBLEMS

(41)

① Solve the difference equation $y_{n+2} - 4y_n = 0$, by using Z-transforms.

→ Taking Z-transform on both sides of the given equation we have,

$$Z_T(y_{n+2}) - 4Z_T(y_n) = Z_T(0)$$

$$\text{i.e., } z^2 [\bar{y}(z) - y_0 - y_1 z^{-1}] - 4\bar{y}(z) = 0$$

$$[z^2 - 4] \bar{y}(z) - 2z = 0, \text{ by using the given values.}$$

$$(a) \quad \bar{y}(z) = \frac{2z}{z^2 - 4}$$

$$\text{Let } \frac{\bar{y}(z)}{z} = \frac{2}{(z-2)(z+2)} = \frac{A}{z-2} + \frac{B}{z+2}$$

$$2 = A(z+2) + B(z-2)$$

$$\text{Put } z=2 \Rightarrow 2 = A(4)$$

$$\therefore \boxed{A = 1/2}$$

$$z = -2 \Rightarrow 2 = B(-4)$$

$$\boxed{B = -1/2}$$

$$\text{Hence, } \bar{y}(z) = \frac{1}{2} \frac{z}{z-2} - \frac{1}{2} \frac{z}{z+2}$$

$$\Rightarrow Z_T^{-1}[\bar{y}(z)] = \frac{1}{2} \left\{ Z_T^{-1} \left[\frac{z}{z-2} \right] - Z_T^{-1} \left[\frac{z}{z+2} \right] \right\}$$

$$\text{i.e., } y_n = \frac{1}{2} \{ 2^n - (-2)^n \} = \frac{2^n}{2} + \frac{(-2)^n}{-2}$$

ie, $z[\bar{y}(z) - y_0] + \frac{1}{4}\bar{y}(z) = \frac{z}{z - \frac{1}{4}}$

$(z + \frac{1}{4})\bar{y}(z) = \frac{z}{z - \frac{1}{4}}$, by using the given value.

$\bar{y}(z) = \frac{z}{(z - \frac{1}{4})(z + \frac{1}{4})}$

Let $\frac{\bar{y}(z)}{z} = \frac{1}{(z - \frac{1}{4})(z + \frac{1}{4})} = \frac{A}{z - \frac{1}{4}} + \frac{B}{z + \frac{1}{4}}$

(*) $1 = A[z + \frac{1}{4}] + B[z - \frac{1}{4}]$

put $z = \frac{1}{4} \Rightarrow 1 = A(\frac{1}{2})$

$\therefore \boxed{A = 2}$

$z = -\frac{1}{4} \Rightarrow 1 = B(-\frac{1}{2})$

$\therefore \boxed{B = -2}$

Hence $\bar{y}(z) = 2 \cdot \frac{z}{z - \frac{1}{4}} - 2 \cdot \frac{z}{z + \frac{1}{4}}$

$\Rightarrow z_T^{-1}[\bar{y}(z)] = 2 \left\{ z_T^{-1} \left[\frac{z}{z - \frac{1}{4}} \right] - z_T^{-1} \left[\frac{z}{z + \frac{1}{4}} \right] \right\}$

Hence, $\boxed{y_n = 2 \left\{ \left[\frac{1}{4} \right]^n - \left[-\frac{1}{4} \right]^n \right\}}$ is the required particular solution.

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Thus, $y_n = 2^{n-1} + (-2)^{n-1}$ is the required particular solution.

⑥ Solve the difference equation $u_{n+2} + u_n = 0$, by using Z-transforms.

→ Taking Z-transform on both sides of given equation we have,

$$Z_T(u_{n+2}) + Z_T(u_n) = Z_T(0)$$

$$\text{i.e., } z^2 [\bar{u}(z) - u_0 - u_1 z^{-1}] + \bar{u}(z) = 0$$

$$[z^2 + 1] \bar{u}(z) = u_0 z^2 + u_1 z$$

$$\bar{u}(z) = u_0 \cdot \frac{z^2}{z^2 + 1} + u_1 \frac{z}{z^2 + 1}$$

$$\Rightarrow Z_T^{-1}[\bar{u}(z)] = u_0 Z_T^{-1}\left[\frac{z^2}{z^2 + 1}\right] + u_1 Z_T^{-1}\left[\frac{z}{z^2 + 1}\right]$$

Thus,
$$u_n = u_0 \cos(n\pi/2) + u_1 \sin(n\pi/2)$$

where u_0 & u_1 are arbitrary constants in the required general solution of the given difference eqn.

③ Solve by using Z-transforms: $y_{n+1} + \frac{1}{4} y_n = \left(\frac{1}{4}\right)^n$
 $(n \geq 0), y_0 = 0$

→ Taking Z-transform on both sides of the given equation we have,

$$Z_T(y_{n+1}) + \frac{1}{4} Z_T(y_n) = Z_T\left[\left(\frac{1}{4}\right)^n\right]$$

③ Solve the difference equation $y_{n+2} + 6y_{n+1} + 9y_n = 2^n$ with $y_0 = y_1 = 0$ using z-transforms. 42

→ Taking z-transform on both sides of the given equation we have,

$$z_T(y_{n+2}) + 6z_T(y_{n+1}) + 9z_T(y_n) = z_T(2^n)$$

$$z^2 [\bar{y}(z) - y_0 - y_1 z^{-1}] + 6z [\bar{y}(z) - y_0] + 9\bar{y}(z) = \frac{z}{z-2}$$

$$(z^2 + 6z + 9) \bar{y}(z) = \frac{z}{z-2}, \text{ by using the initial values.}$$

$$\bar{y}(z) = \frac{z}{(z-2)(z+3)^2}$$

$$\text{Let } \frac{z}{(z-2)(z+3)^2} = A \cdot \frac{z}{z-2} + B \cdot \frac{z}{z+3} + C \cdot \frac{z}{(z+3)^2}$$

$$1 = A(z+3)^2 + B(z-2)(z+3) + C(z-2)$$

$$\text{put } z = 2 \Rightarrow 1 = A(25) \Rightarrow \boxed{A = 1/25}$$

$$z = -3 \Rightarrow 1 = C(-5) \Rightarrow \boxed{C = -1/5}$$

Equating the co-efficient of z^2 on both sides we get.

$$0 = A + B \Rightarrow \boxed{B = -1/25}$$

$$\text{Hence, } \bar{y}(z) = \frac{1}{25} \cdot \frac{z}{z-2} - \frac{1}{25} \cdot \frac{z}{z+3} - \frac{1}{5} \cdot \frac{z}{(z+3)^2}$$

$$\bar{y}(z) = \frac{1}{25} \cdot \frac{z}{z-2} - \frac{1}{25} \cdot \frac{z}{z+3} - \frac{1}{5} \cdot \frac{1}{-3} \cdot \frac{-3z}{(z+3)^2}$$

$$\Rightarrow Z_T^{-1}[\bar{y}(z)] = \frac{1}{25} Z_T^{-1}\left[\frac{z}{z-2}\right] - \frac{1}{25} Z_T^{-1}\left[\frac{z}{z+3}\right] + \frac{1}{15} Z_T^{-1}\left[\frac{-3z}{(z+3)^2}\right]$$

$$\therefore y_n = \frac{1}{25} (2)^n - \frac{1}{25} (-3)^n + \frac{1}{15} (-3)^n \cdot n$$

Thus, $y_n = \frac{1}{5} \left\{ \frac{1}{5} (2)^n - \frac{1}{5} (-3)^n + \frac{1}{3} (-3)^n \cdot n \right\}$ is the required solution.

⑤. Solve by using Z-transform $y_{n+2} + 2y_{n+1} + y_n = n$
with $y_0 = 0 = y_1$

→ Taking Z-transform on both sides of the given eqn we have,

$$Z_T(y_{n+2}) + 2Z_T(y_{n+1}) + Z_T(y_n) = Z_T(n)$$

$$z^2 [\bar{y}(z) - y_0 - y_1 z^{-1}] + 2z [\bar{y}(z) - y_0] + \bar{y}(z) = \frac{z}{(z-1)^2}$$

$$\text{i.e., } [z^2 + 2z + 1] \bar{y}(z) = \frac{z}{(z-1)^2} \text{ , by using the given values.}$$

$$\bar{y}(z) = \frac{z}{(z-1)^2(z+1)^2}$$

$$\text{Let, } \frac{z}{(z-1)^2(z+1)^2} = A \cdot \frac{z}{z-1} + B \cdot \frac{z}{(z-1)^2} + C \cdot \frac{z}{z+1} + D \cdot \frac{z}{(z+1)^2} \quad \text{--- (1)}$$

$$1 = A(z-1)(z+1)^2 + B(z+1)^2 + C(z-1)^2(z+1) + D(z-1)^2$$

$$\text{put } z = 1 \Rightarrow 1 = B(4) \therefore \boxed{B = 1/4}$$

$$z = -1 \Rightarrow 1 = D(4) \therefore \boxed{D = 1/4}$$

Equating the co-efficient of z^3 on both sides we get,

$$A + C = 0 \quad (1) \quad C = -A$$

put $z = 0 \Rightarrow 1 = -A + B + C + D$

i.e., $1 = C + \frac{1}{4} + C + \frac{1}{4}$

$$\frac{1}{2} = 2C$$

$$\therefore C = \frac{1}{4}$$

Also $A = -\frac{1}{4}$

Substituting $A = -\frac{1}{4}$, $B = C = D = \frac{1}{4}$ in (1) & taking the inverse z -transform we have,

$$\begin{aligned} Z_T^{-1} [\bar{y}(z)] &= -\frac{1}{4} Z_T^{-1} \left[\frac{z}{z-1} \right] + \frac{1}{4} Z_T^{-1} \left[\frac{z}{(z-1)^2} \right] \\ &\quad + \frac{1}{4} Z_T^{-1} \left[\frac{z}{z+1} \right] + \frac{1}{4} Z_T^{-1} \left[\frac{z}{(z+1)^2} \right] \end{aligned}$$

$$\begin{aligned} \text{i.e., } y_n &= -\frac{1}{4} \cdot 1 + \frac{1}{4} n + \frac{1}{4} (-1)^n + \frac{1}{4} \cdot (-1) (-1)^n n \\ &= \frac{1}{4} \{ (n-1) - (-1)^n (n-1) \} \end{aligned}$$

Thus, $y_n = \frac{(n-1)}{4} [1 - (-1)^n]$ is the required solution.